

```
exercise_5.py  x
1  ## Course 1: Natural Language Processing
2  ## Module 1: Intro to NLP in Python
3  ## Coding Exercise 5
4  import nltk
5
6  def remove_verbs(sents):
7      sent = ' '.join(sents)
8      sent = nltk.word_tokenize(sent)
9      sent = nltk.pos_tag(sent)
10     pattern = r""""ChinkAllVerbs: {<.*>+}
11     }<VB.*>{""""  #'NP: {<DT>?<JJ>*<NN>}'
12     cp = nltk.RegexpParser(pattern)
13     cs = cp.parse(sent)
14     return cs
15
16 sentences = ["He washed the car yesterday.", "I bought her a book and a coffee.", "She thinks
17 print(remove_verbs(sentences))
18
```

Collapse

Coding Exercise 5

Coding Exercise 5

Create the function `remove_verbs` that takes a list of strings and returns all parts of speech which are chunked together except for verbs.

Expected Output

After creating the `remove_verbs` function, pass it the following variable and print the return value.

```
sentences = ["He washed the car yesterday.", "I bought her a book and a coffee.", "
```

Click the `TRY IT` button to run the file and check your work. The expected output is shown below.

TRY IT

```
(S
  (ChinkAllVerbs He/PRP)
  washed/VBD
  (ChinkAllVerbs the/DT car/NN yesterday/NN ./
I/PRP)
  bought/VBD
  (ChinkAllVerbs
    her/PRP
    a/DT
    book/NN
    and/CC
    a/DT
    coffee/NN
    ./
    She/PRP)
  thinks/VBZ
  going/VBG
  (ChinkAllVerbs outside/JJ)
  is/VBZ
  (ChinkAllVerbs healthy/JJ ./ The/DT dog/NN)
  runs/VBZ
  (ChinkAllVerbs away/RB ./))
```

```
(S
  (ChinkAllVerbs He/PRP)
  washed/VBD
  (ChinkAllVerbs the/DT car/NN yesterday/NN ./ I/PRP)
  bought/VBD
  (ChinkAllVerbs her/PRP a/DT book/NN and/CC a/DT coffee/NN ./ She/PRP)
  thinks/VBZ
  going/VBG
  (ChinkAllVerbs outside/JJ)
  is/VBZ
  (ChinkAllVerbs healthy/JJ ./ The/DT dog/NN)
  runs/VBZ
  (ChinkAllVerbs away/RB ./))
```

Submit your work for evaluation.

Here is one possible solution:

```
from nltk.corpus import reuters
from nltk.tokenize import word_tokenize
from nltk import pos_tag, RegexpParser

sentences = ["He washed the car yesterday.", "I bought her a book and a

def remove_verbs(data):
    grammar = r""""ChinkAllVerbs: {<.*>+}
    }<VB.*>{""""
    parser = RegexpParser(grammar)
    for sentence in data:
        tokens_by_sentence = [word_tokenize(sentence) for sentence in data]
        tokens = sum(tokens_by_sentence, [])
        tags = pos_tag(tokens)
        chunked_data = parser.parse(tags)
        return(chunked_data)

    print(remove_verbs(sentences))
```

- Import `reuters`, `word_tokenize`, `pos_tag`, and `RegexpParser` from the `nltk` module
- Create the `remove_words` function that takes a list of strings
- Create a grammar that chunks everything except for verbs
- Apply `RegexpParser` to the grammar
- Iterate over the list of strings
- Tokenize each sentence
- Flatten the 2D list of tokens
- Apply the parts of speech tags to the tokens
- Parse all of the tagged data
- Return the chunked data

Check It!

```
LAST RUN on 12/27/2023, 4:22:24 AM

(S
  (ChinkAllVerbs He/PRP)
  washed/VBD
  (ChinkAllVerbs the/DT car/NN
yesterday/NN ./ I/PRP)
  bought/VBD
  (ChinkAllVerbs
    her/PRP
    a/DT
    book/NN
```

Mark as Completed

Back to dashboard